Lab 1: Understanding Network Equipment, Wiring, and Cross-Over Testing

Objective

The goal of this lab is to get a better understanding of the different types of network equipment, how they're connected using cables, and how to test those connections especially using cross-over wiring. This knowledge is essential for anyone working with computer networks.

Theory

When we talk about computer networks, we often focus on the software—but just as important is the physical hardware that makes everything work. These pieces of equipment help data move smoothly from one device to another. Here’s a quick look at the key devices:

1. Router

A router acts like a traffic director for data. It decides the best path for data to travel from one network to another, like between your home network and the internet. Routers work at Layer 3 (the Network layer) of the OSI model, using routing tables to make their decisions.

2. Switch

A switch is used to connect multiple devices in the same network, like computers in an office. It sends data only to the specific device it’s meant for, which makes the network faster and more efficient. Switches operate at Layer 2 (the Data Link layer) and use MAC addresses to know where to send the data.

3. Hub

Hubs are simpler than switches. When a hub gets data, it sends it to every device connected to it, whether they need it or not. That makes them less efficient and mostly outdated, but they still help us understand how networks evolved. Hubs work at Layer 1 (Physical layer).

4. Network Interface Card (NIC)

Every device that connects to a network needs a NIC. It's the hardware that allows computers to send and receive data either through a cable (Ethernet) or wirelessly. Each NIC has its own unique MAC address, which helps identify devices on the network.

5. Firewall

Think of a firewall as a security guard for your network. It controls which data is allowed in or out based on specific rules. Firewalls can be hardware or software, and they help protect against threats like hackers or malware.

6. Access Point (AP)

Access points allow wireless devices like smartphones or laptops to connect to a wired network. They help extend Wi-Fi coverage and are especially useful in larger spaces. They operate at both the physical and data link layers of the OSI model.

Conclusion

Learning about network equipment and how it all connects is a critical step in building, managing, and securing a computer network. Each device plays its own role, from directing traffic to securing data and providing wireless access. Knowing how to wire these devices properly especially using tools like cross-over cables and being able to test those connections gives you the foundation to troubleshoot and build reliable network systems.

Lab 2: Introduction to packet tracer simulation tools.

Objective:

To learn how to set up a basic network using Cisco Packet Tracer and understand how devices like routers, switches, and PCs communicate.

Theory:

Packet Tracer is a tool made by Cisco that lets you build and test virtual networks. It's great for practicing without needing real equipment. You can connect devices, assign IPs, and see how data moves through the network.

In this lab, we worked with Cisco Packet Tracer, which is a simulation tool designed to help us build and test virtual networks. It's commonly used in networking courses because it lets you practice setting up routers, switches, and computers without needing any real hardware.

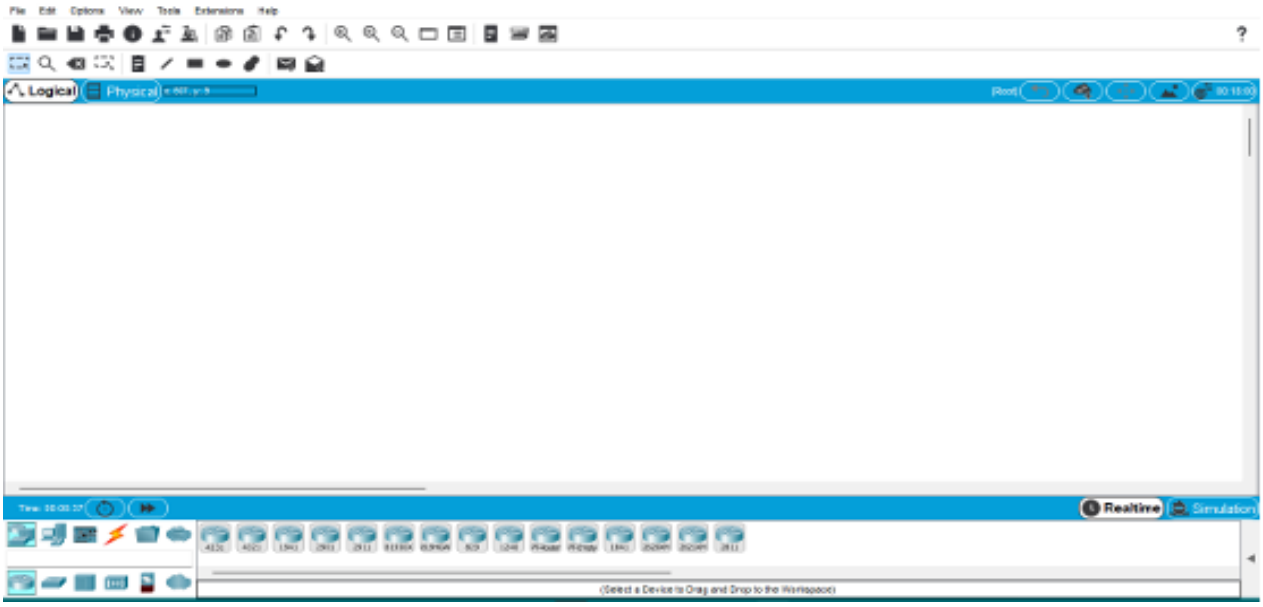


Figure 1: Interface of Cisco Packet Tracer

Observation and Find: Testing the connectivity between Routers, Switch, PC, Laptop and Server.

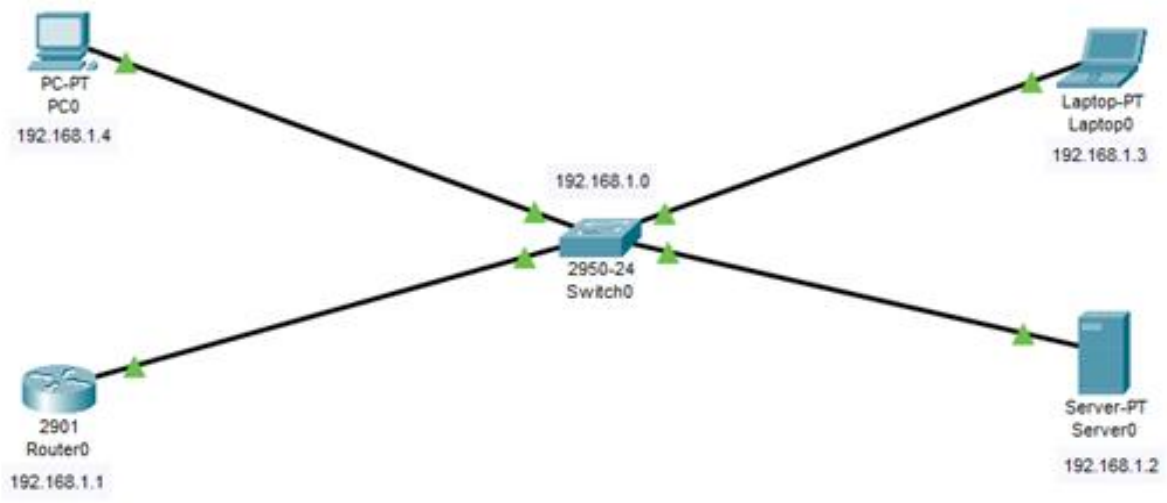


Figure 2: Interface of Cisco packet tracker

Output:

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=305ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 305ms, Average = 76ms
```

Conclusion:

This lab helped us get comfortable with Cisco Packet Tracer and showed us the basics of how networks are built and tested. It was a good starting point for learning networking hands-on.

Lab 3: Creating and configuring a simple peer-to-peer network having two PCs and testing the connectivity between them.

Objective:

1. Set up a peer-to-peer network using Cisco Packet Tracer
2. Test the connection between two PCs
3. Use Command Prompt to ping one PC from the other and check for successful communication

Theory:

A peer-to-peer (P2P) network allows two computers to communicate directly without a central device like a router or switch. In this lab, we connect two PCs using an Ethernet cable, assign them IP addresses in the same range, and test connectivity using the ping command. If the ping is successful, it means the network setup is correct.

Observation and Find:

Implementing peer-to-peer network in between two PCs and testing connectivity between them.

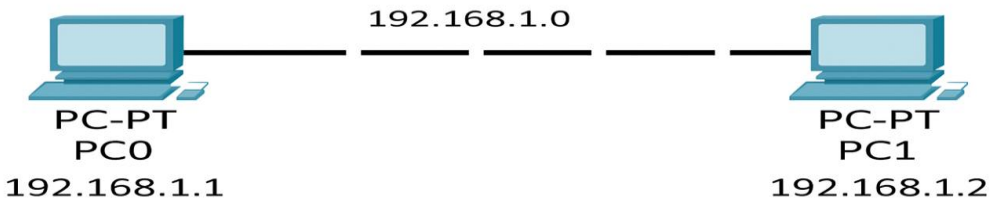


Figure 3: peer to peer network

Output:

The ping command returned replies from the opposite PC, proving that the network configuration was correct and the two computers could communicate directly.

Discussion:

This lab showed how two computers can directly communicate without using any network device. It helped us understand basic IP addressing, simple cabling, and how to use the command prompt for connectivity testing.

Conclusion:

We learned how to create a basic peer-to-peer connection between two PCs using Packet Tracer and verified connectivity using the ping command. This exercise gave us a clear idea of how simple networks function and communicate.

Lab 4: Creating a local area network and testing the connectivity within the network.

Objectives:

1. Build a basic LAN using Cisco Packet Tracer
2. Test connectivity between devices within the network
3. Understand how LANs work and where they're used

Theory:

A Local Area Network (LAN) connects computers and other devices within a limited area like a home, school, or office. It allows for fast, reliable communication, file sharing, and access to shared resources like printers or servers.

LANs typically use Ethernet cables and switches to connect devices. Each device gets an IP address so they can talk to each other. LANs can also include Wi-Fi for wireless connections.

Key features of LANs:

- Fast data speeds (10 Mbps – several Gbps)
- Low delay and high reliability
- Limited range (usually one building or room)
- Easy to set up and manage

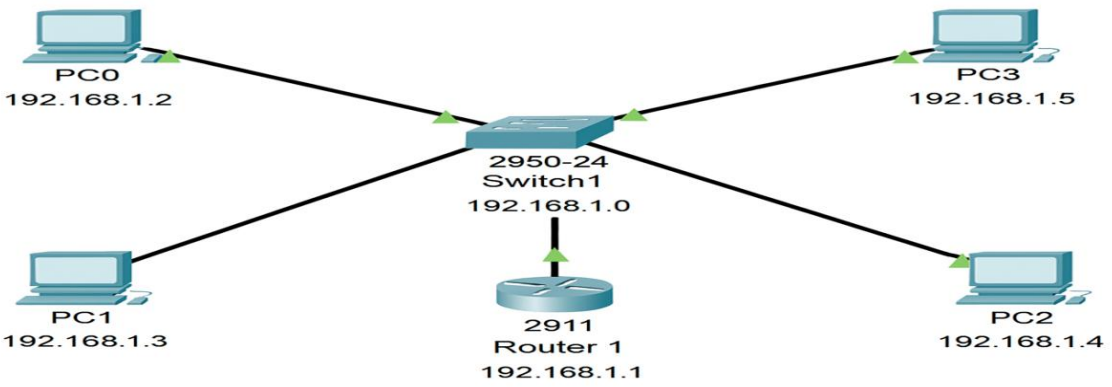


Figure 4: Local Area Network

Output: After assigning IP addresses to all PCs and connecting them through a switch:

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Discussion:

This lab gave us hands-on experience in building and testing a basic LAN. By assigning IPs and connecting devices through a switch, we saw how communication happens in a local network. The successful ping results proved that the setup was correct and functional.

Conclusion:

We successfully created a working LAN and confirmed that all connected devices could communicate. This lab helped us understand how LANs operate and their importance in real-world network setups.

Lab 5: Interconnecting two different LANs and testing the connectivity between them

Objectives:

1. Connect two separate LANs using a router
2. Test data communication between the two LANs

Theory:

In real-world networks, it's often necessary to connect two different LANs to allow communication between them. This process is called LAN interconnection or internetworking, and it's typically done using a router.

In this lab, we created two separate LANs, each with its own set of devices connected through a switch. A router was used to connect the LANs and enable communication. This kind of setup is common in businesses, schools, or any place with multiple departments needing network access.

Observation and Find:

Successfully created two LANs, assigned PCs into different LANs, and verified communication by pinging devices within the same LAN and across LANs (after router connection).

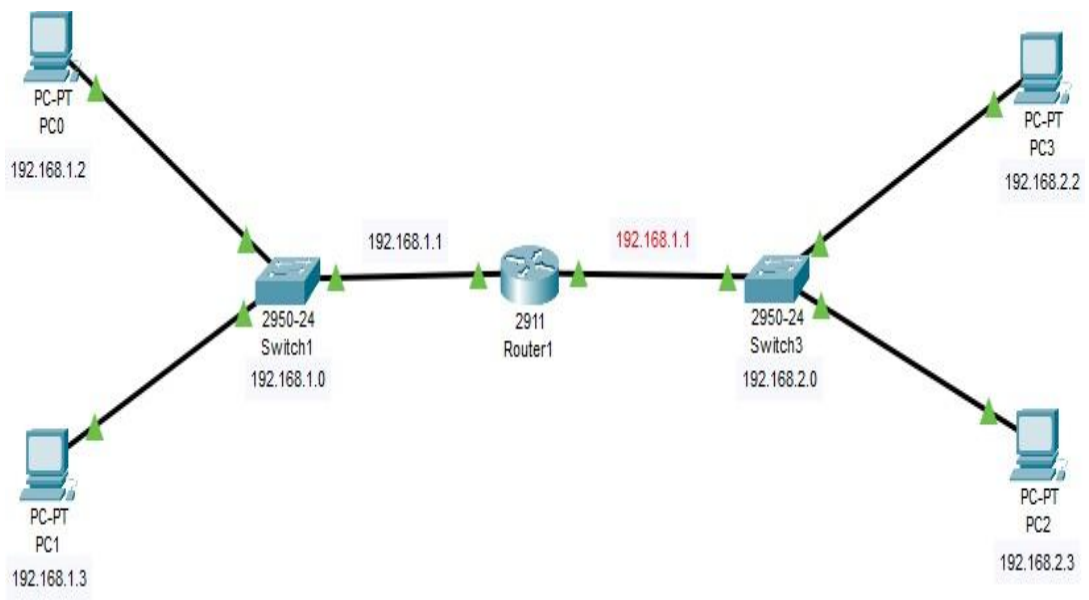


Figure 1 : Interconnection of Local Area Network

Output:

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time=1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Discussion:

This lab showed how routers help connect separate LANs and allow data to travel between them. It also reinforced how IP addressing and subnetting work in a multi-network environment. The successful ping results confirmed proper internetwork configuration.

Conclusion:

The lab helped us understand how to connect two LANs using a router and test their communication. It gave us hands-on experience with internetworking and the role routers play in enabling communication between different networks.

Lab 6: Router configuration using command line interface.

Objectives:

1. Understand how to configure a router manually
2. Learn key commands used in the router's Command Line Interface (CLI)

Theory:

A **router** is a device that connects different networks and directs data between them. To function properly, routers need to be manually configured with IP addresses, subnet masks, and sometimes security settings.

This is typically done through the Command Line Interface (CLI), where we enter a series of commands to set up the router. CLI gives network administrators full control and is a core part of Cisco networking.

Process for Router Configuration using CLI:

1. Enter privileged mode by typing `enable` or `en`.
2. Access the console line using `line console 0`.
3. Set a console password for secure access.
4. Exit back to the main mode.
5. Re-enter using the password to check authentication.
6. Enter global configuration mode with `configure terminal` (or `conf t`).
7. Assign IP address and subnet mask to the router's GigabitEthernet interface using:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#ip address 192.168.0.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int fa 0/1
Router(config-if)#ip address 172.16.0.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#exit
```

```
Router#show running-conf
Building configuration...

Current configuration : 578 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
```

```
Router#copy running-conf startup-conf
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
```

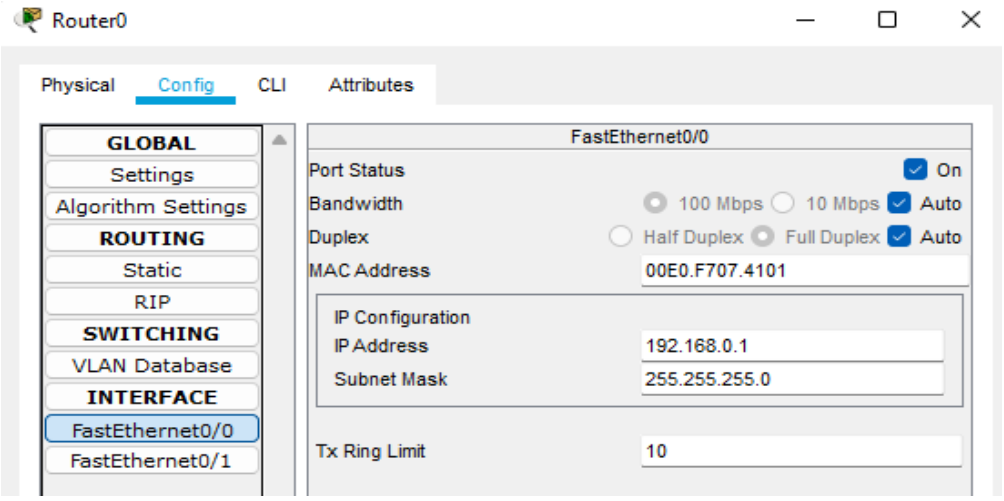


Figure 2 : IP Configuration of Router

Observation and Finding: Configuring routers using command Line Interface.

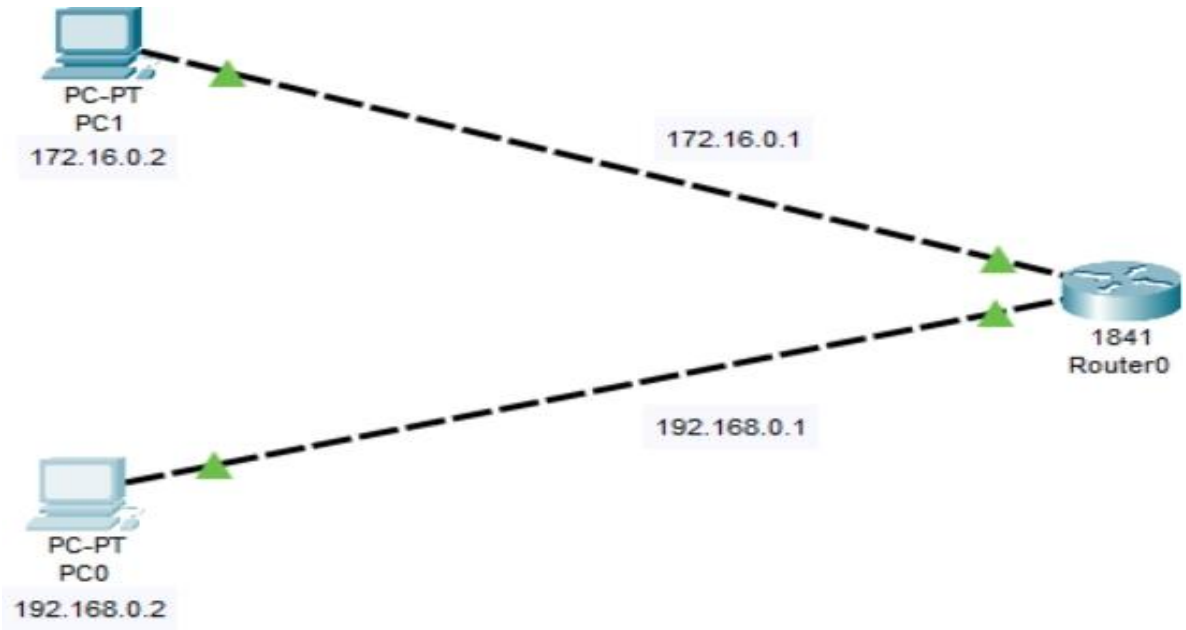


Figure 3 : Configuration of a Router

Output:

```
C:\>ping 172.16.0.2

Pinging 172.16.0.2 with 32 bytes of data:

Reply from 172.16.0.2: bytes=32 time<1ms TTL=127
Reply from 172.16.0.2: bytes=32 time<1ms TTL=127
Reply from 172.16.0.2: bytes=32 time<1ms TTL=127
Reply from 172.16.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Discussion: Configuration of a router using Command Line Interface.

Conclusion: The aim of the lab is to become familiar with Command Line Interface

Lab 7: Static Routing

Objectives:

1. Understand the concept of static routing
2. Learn how to configure static routes on a router using IP route commands

Theory:

Static routing is a method where routes are manually configured on a router by a network administrator. Unlike dynamic routing, static routes don’t change automatically they are fixed paths that the router uses to send data between networks.

Static routing is useful in smaller or more controlled networks where routing paths don’t change frequently. It's simple, efficient, and offers better control, but requires manual updates if the network changes.

Steps for Static Routing Configuration:

1. Assign IP addresses to interfaces on Router 1
2. Connect PCs and switches to Router 1’s network
3. Do the same for Router 2 (assign IPs, connect to devices)
4. Enable router interfaces using no shutdown
5. Configure static routes using the ip route command on both routers:
6. ip route [destination_network] [subnet_mask] [next_hop_ip]
7. Test end-to-end connectivity using the ping command between PCs on different networks

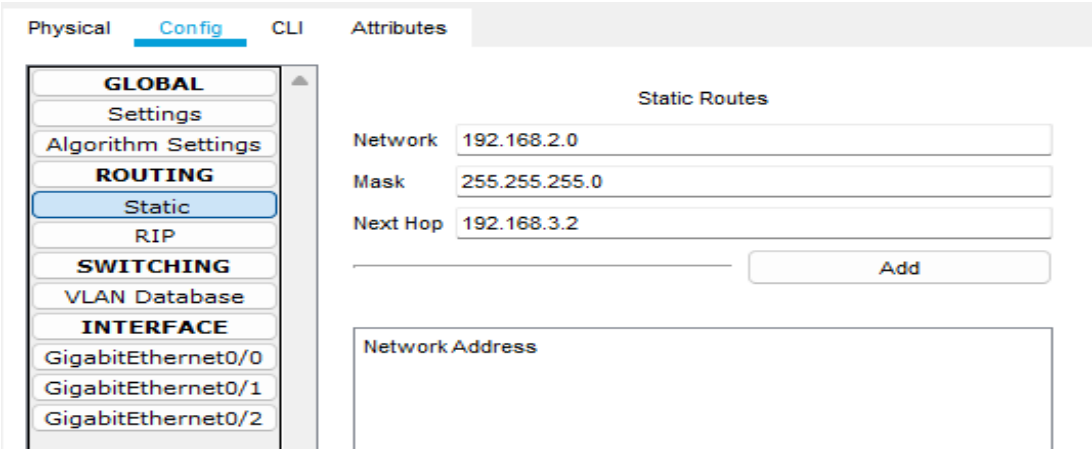


Figure 4 : Static Routes Network

Observation:

We manually added static routes on both routers. After configuring IPs and routing tables, we successfully established communication between devices on different LANs.

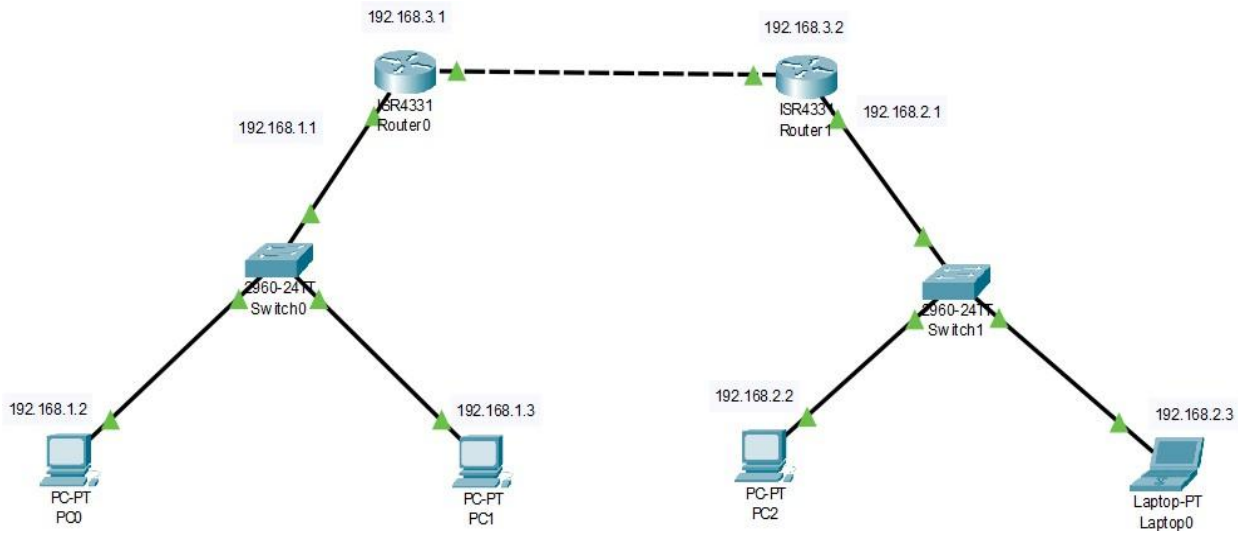


Figure 9: Static routes for static routing

Output:

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Discussion:

In this lab, we practiced both GUI and CLI methods to implement static routing. It showed us how routers forward packets using manually defined paths and how important correct IP configuration and route entries are for successful communication.

Conclusion:

The lab successfully demonstrated how to configure and test static routing. We learned the key commands and saw how routers use static paths to send data between different networks.

Lab 8: Implementing Dynamic Routing Protocol (RIP)

Objectives:

1. Understand how dynamic routing works
2. Learn to configure the RIP (Routing Information Protocol) on routers

Theory:

Dynamic routing allows routers to automatically learn and update routes by exchanging information with other routers. One of the simplest dynamic protocols is RIP (Routing Information Protocol), which uses hop count to determine the best path to a destination.

In RIP:

- Each router shares its routing table with neighboring routers
- The maximum hop count allowed is 15 (16 means unreachable)
- RIP helps reduce manual configuration and adapts to network changes automatically

Process for RIP Routing:

1. Configure at least two routers in the network topology.
2. Connect switches and desktops with proper IP addresses for communication.
3. Enable the interfaces of each router using the `no shutdown` command.
4. Enable RIP on the routers using CLI with the following commands:

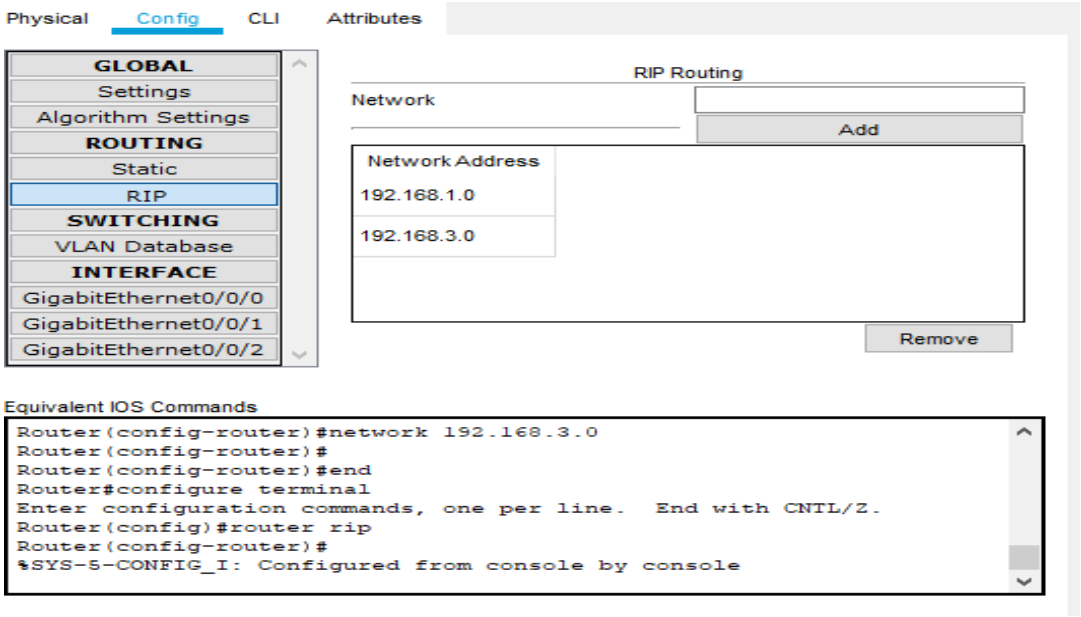


Figure 5 : RIP network routes

Observation:

After configuring RIP on both routers and assigning correct network addresses, we observed that the routers automatically exchanged route information. Devices on different networks were able to communicate successfully.

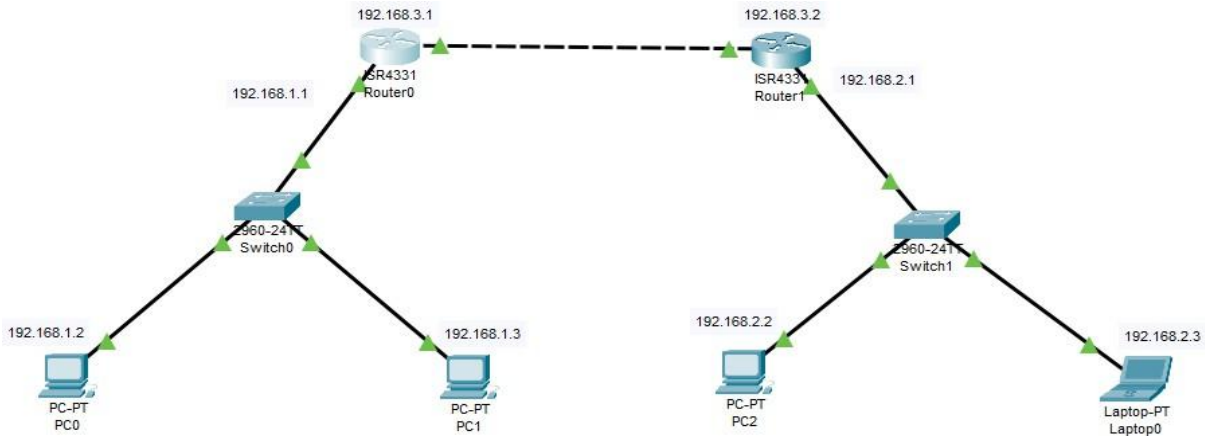


Figure 6 : Dynamic Routing (RIP)

Output:

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time<1ms TTL=254
Reply from 192.168.2.1: bytes=32 time<1ms TTL=254
Reply from 192.168.2.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Discussion:

This lab introduced RIP, one of the basic dynamic routing protocols. By setting up RIP on routers, we saw how routing tables are updated automatically, without the need for manual route entries like in static routing.

Conclusion:

The lab helped us understand how RIP works and how to implement it using Cisco Packet Tracer. We successfully built a dynamic network where routers learned and shared routing information automatically.

Lab 9: Configuring DNS server to assign IP address dynamically

Objectives:

1. To understand the role of a DNS/DHCP server in a network.
2. To learn how a DHCP server assigns IP addresses dynamically to clients.

Theory:

Dynamic Host Configuration Protocol (DHCP) is a client–server protocol that automatically assigns IP addresses and other configuration parameters (such as subnet mask, default gateway, and DNS server) to hosts in a network. This reduces the need for manual IP configuration and ensures that devices can quickly join and communicate in a network.

DHCP is essential in medium to large networks, as it simplifies network administration by dynamically leasing IP addresses to hosts. Along with DHCP, the Domain Name System (DNS) helps resolve hostnames into IP addresses, allowing devices to communicate using user-friendly names instead of numerical IP addresses.

By configuring a DHCP server, network devices (desktops, laptops, etc.) can automatically obtain their IP addresses and connect seamlessly to the network.

Process for DHCP Server to Assign IP Address:

1. Configure a router and connect it with one server and a desktop in the topology.
2. Assign IP addresses to both the router and the server, and enable the router ports using the no shutdown command.
3. Open the server configuration window, go to the Services tab, and enable the DHCP service.
4. Set the default gateway (router IP) and define the starting IP address pool for clients.
5. On the desktop, go to the IP configuration section and select DHCP instead of static IP.
6. The desktop will automatically receive an IP address from the DHCP server.

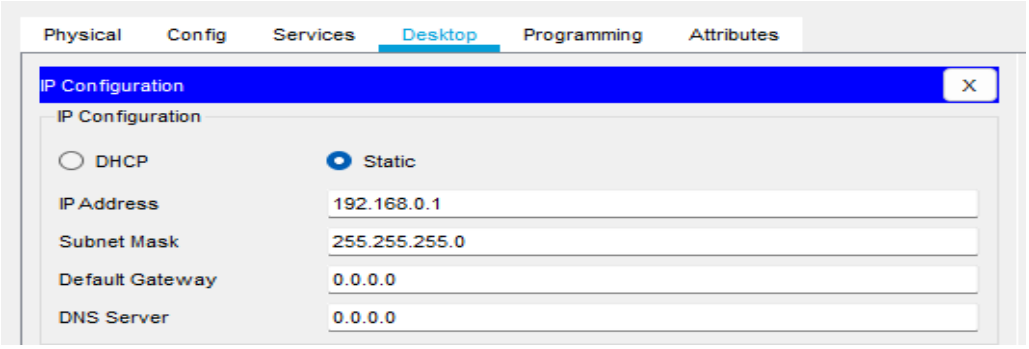


Figure 7 : IP Configuration of Server (DHCP)

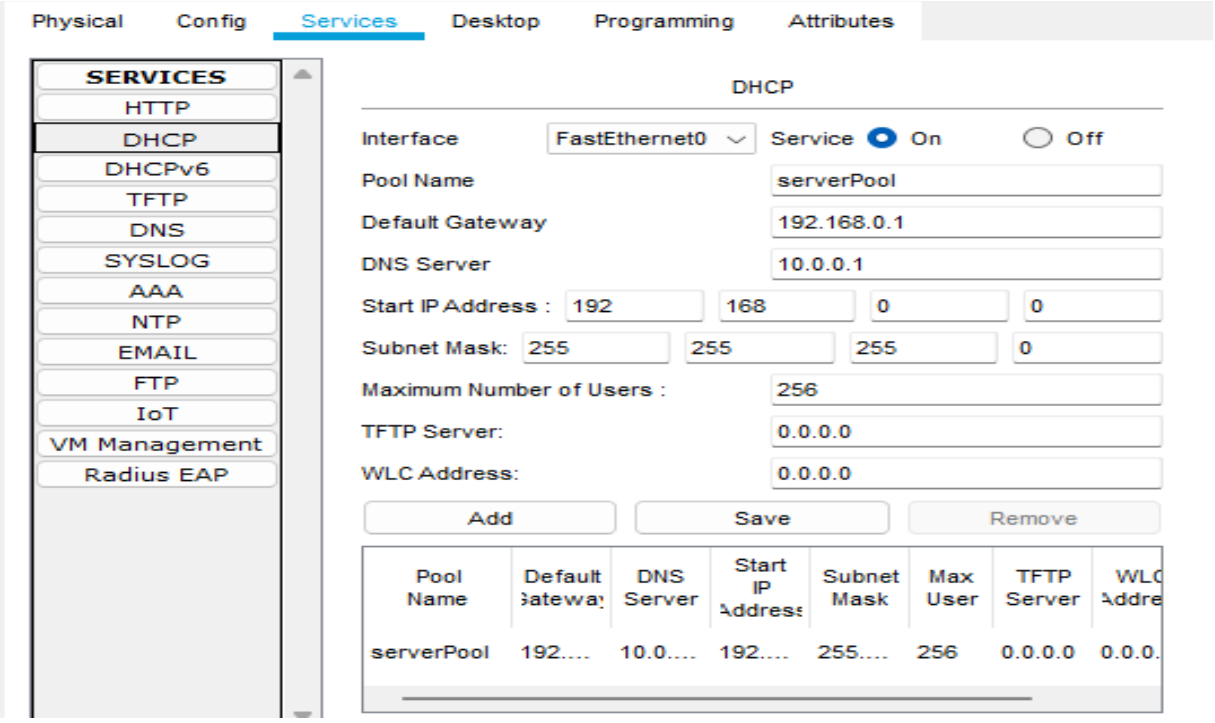


Figure 8 : Server Configuration (DHCP)

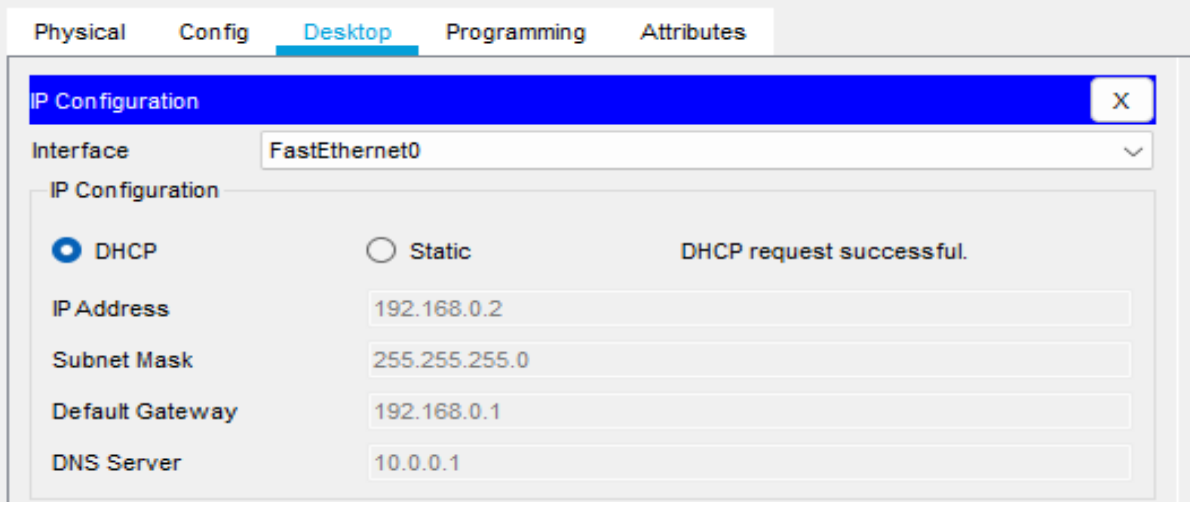


Figure 9 : IP assignment Dynamic (DHCP)

Observation and Findings:

- DHCP service successfully assigned IP addresses to the client machine.
- The router acted as a default gateway for the client.
- The process eliminated the need for manual IP configuration.

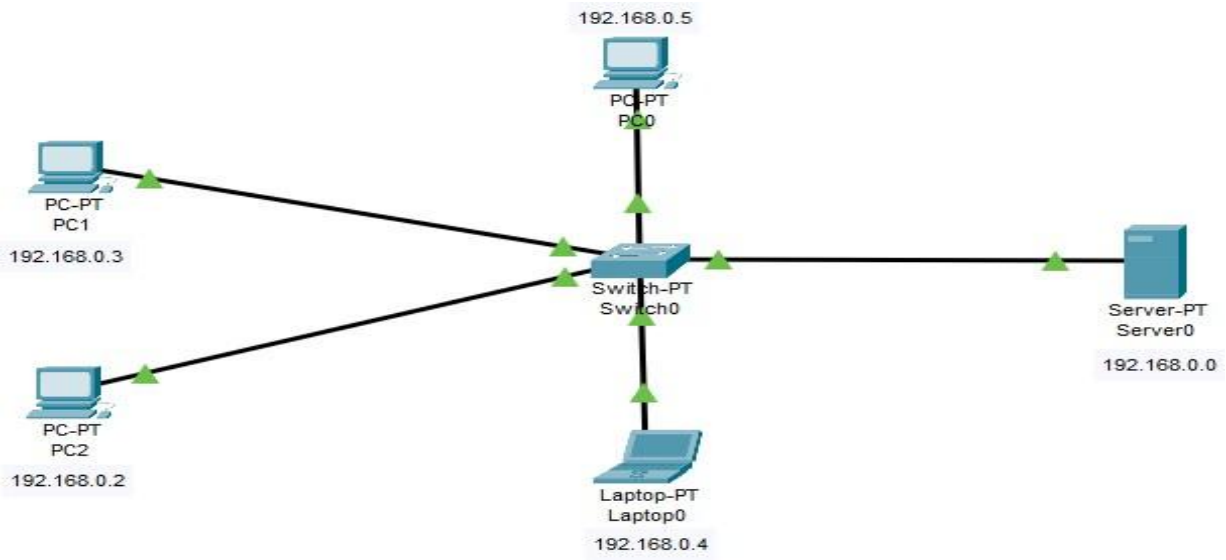
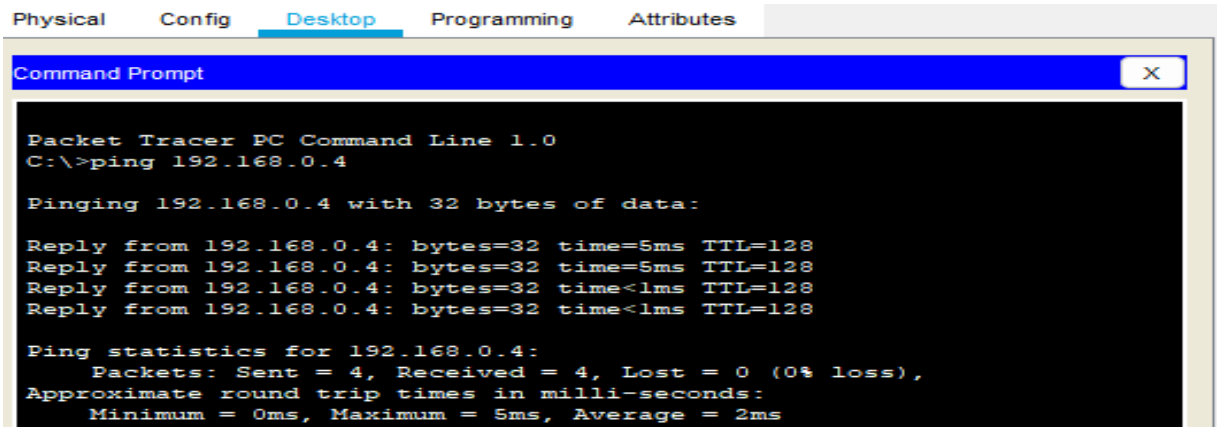


Figure 10 : DHCP Server configuration (DHCP)

Output:



Discussion:

Configuring a DHCP server is crucial in modern networks, as it automates the assignment of IP addresses and reduces administrative effort. In this lab, DHCP assigned IP addresses dynamically to a client, and the router ensured connectivity. The DNS server, when integrated, allows resolution of hostnames to IP addresses, enabling smoother communication across networks.

Conclusion:

The aim of this lab was successfully achieved. DHCP server configuration was performed, and dynamic IP assignment to client devices was verified. This provided practical understanding of how DHCP simplifies network management and ensures efficient communication in an IP network.

Lab 10: Capturing and Analyzing Packets Using Wireshark

Objectives:

1. Learn what Wireshark is and what it's used for
2. Get familiar with the Wireshark interface and how to capture live network traffic

Theory:

Wireshark is a free, open-source tool used to capture and analyze network packets. It’s widely used in networking, cybersecurity, and IT education to monitor data flow, debug communication issues, and understand how different network protocols work.

Wireshark supports cross-platform usage, meaning it runs on operating systems like Windows, Linux, macOS, BSD, and others. It uses pcap (packet capture) to record network traffic from selected interfaces (like Wi-Fi or Ethernet). Each captured packet can be viewed in detail, showing headers and payloads from different layers (like Ethernet, IP, TCP/UDP, etc.).

Steps to Capture Packets in Wireshark:

1. Open Wireshark
2. Select a live network interface (e.g., Wi-Fi)
3. Click the interface to start capturing packets
4. Observe the real-time packet flow
5. Click on any packet to view header details like source IP, destination IP, protocol type, etc.

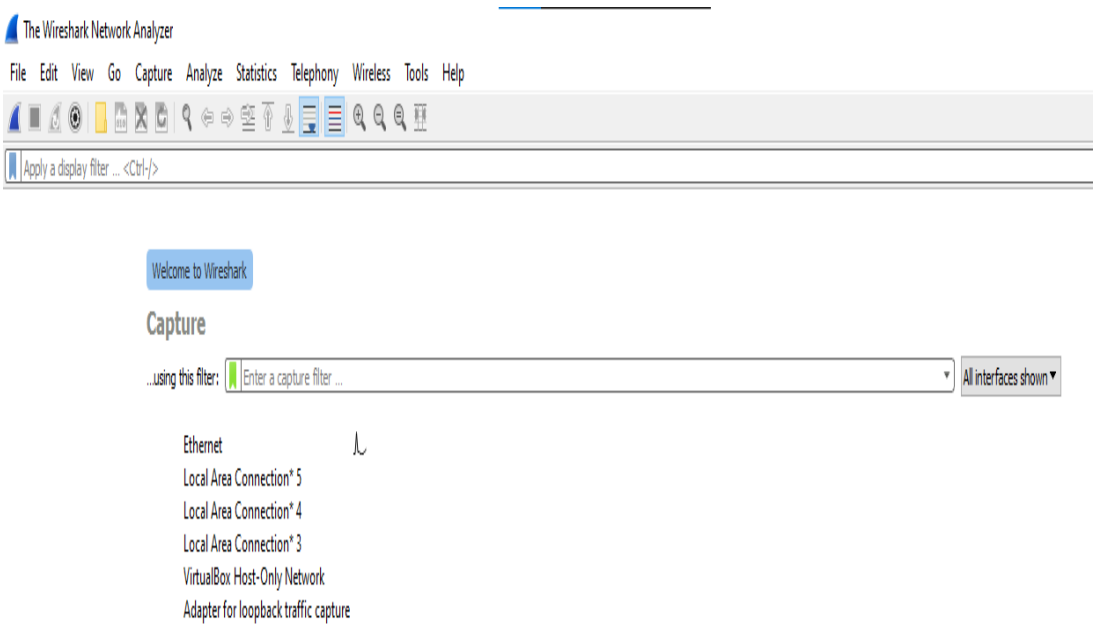


Figure 11 : Wireshark Interface

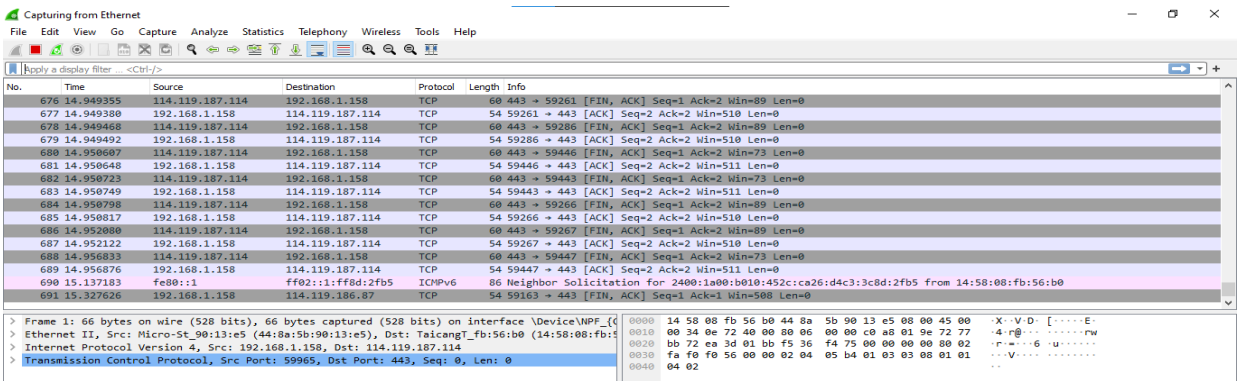


Figure 12 : Flow of packet in Wireshark

Discussion:

This lab introduced us to the basics of Wireshark and how it's used in network analysis. By capturing live traffic, we could clearly see how packets are structured and what information is included in each layer. This is a valuable tool for anyone working in networking or cybersecurity.

Conclusion:

We successfully used Wireshark to capture and analyze live network packets. This lab helped us understand how data travels through a network, what packet headers contain, and how to use Wireshark for real-time traffic monitoring and analysis.